Question	Answer	Marks
10(a)	two from: • frequency below which electrons not ejected • <u>maximum</u> energy of electron depends on frequency • <u>maximum</u> energy of electrons does not depend on intensity • instantaneous emission of electrons	**B2**
10(b)(i)	(λ_0 is the) threshold wavelength **or** wavelength corresponding to threshold frequency **or** maximum wavelength for emission of electrons	**B1**
10(b)(ii)1.	intercept = $1/\lambda_0 = 2.2 \times 10^6\,\text{m}^{-1}$ $\lambda_0 = 4.5 \times 10^{-7}$ m or 450 nm	**A1**
10(b)(ii)2.	gradient = hc	**C1**
	gradient = 2.0×10^{-25} or correct substitution into gradient formula	**C1**
	$h = (2.0 \times 10^{-25}) / (3.0 \times 10^8) = 6.7 \times 10^{-34}$ J s	**A1**
10(c)	line: same gradient	**B1**
	straight line, positive gradient, intercept at greater than 2.2×10^6 when candidate's line extrapolated	**B1**